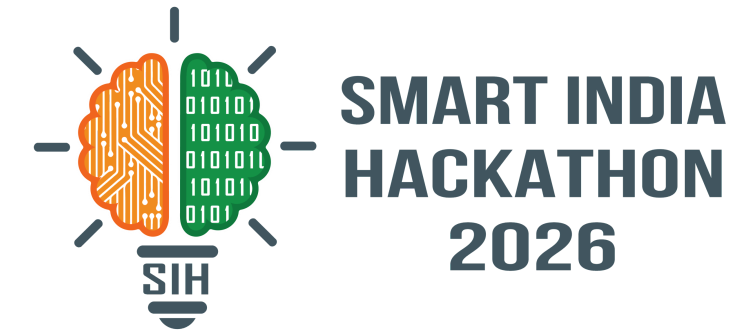


SMART INDIA HACKATHON 2026



- **Problem Statement ID - SIH26006**
- **Problem Statement Title – Intelligent Freight Forecasting for Optimized Vessel Chartering & Bulk Cargo Procurement**
- **Theme - Transportation and logistics**
- **PS Category - Software**
- **Team ID-**
- **Team Name - RouteX**



Intelligent Freight Forecasting and Vessel Optimization

What's Broken Today?

01 Freight decisions are reactive

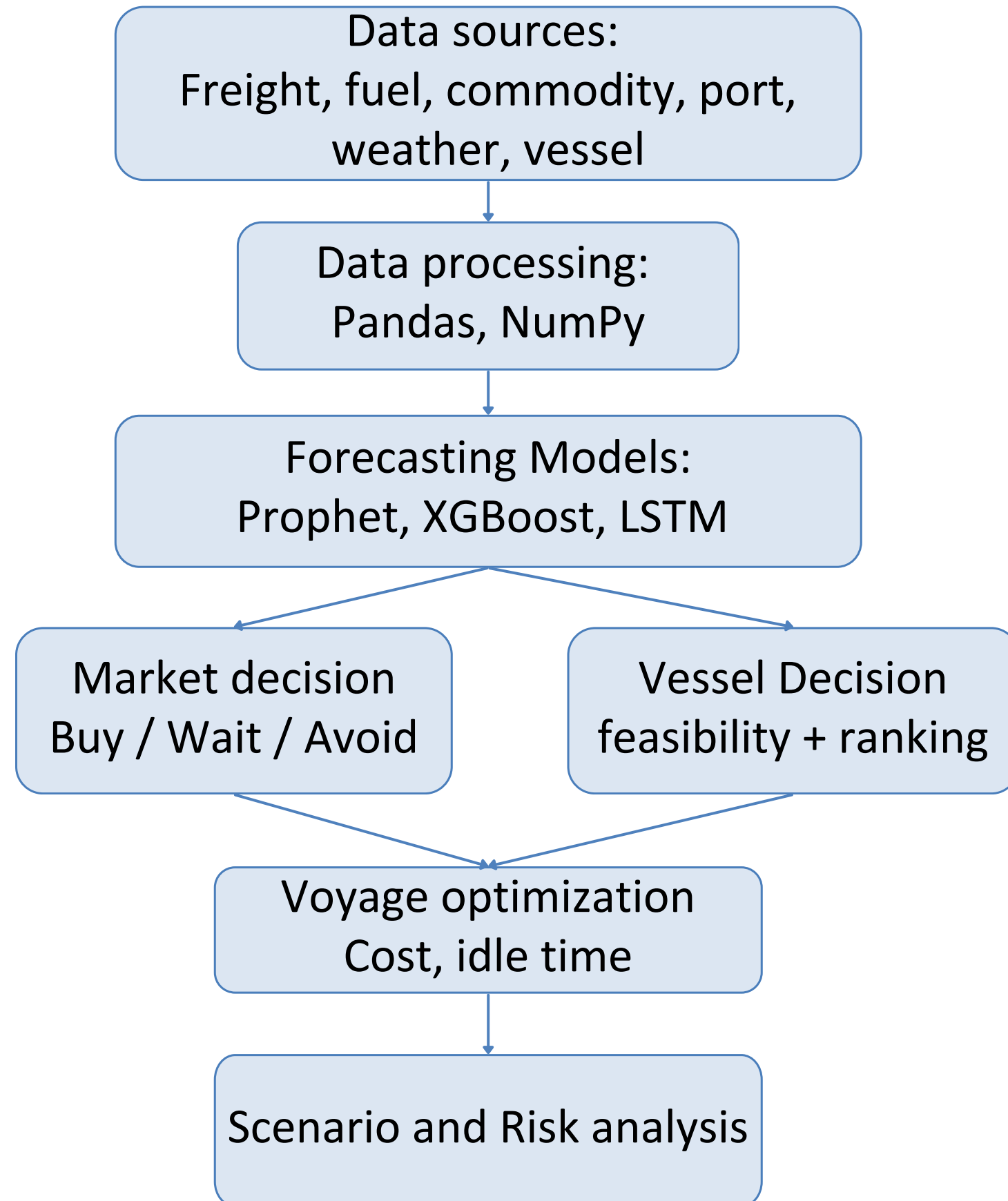
02 Vessel selection is fragmented

03 Voyages are optimized in isolation

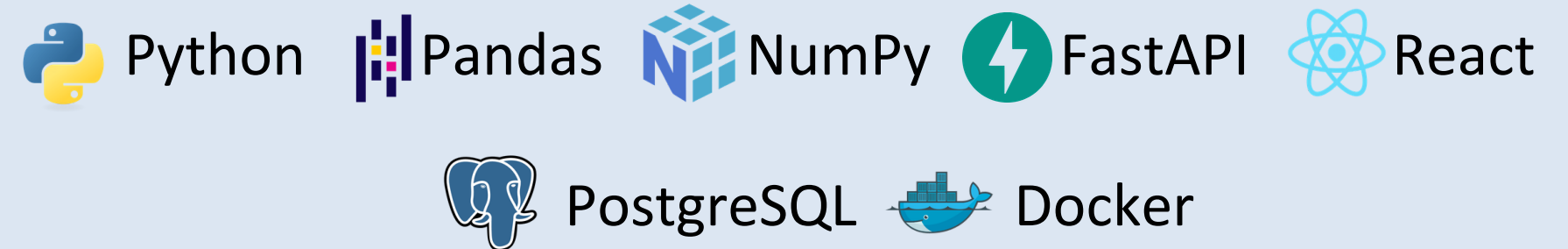
04 "What-if" risks are hard to quantify

Proposed Solution:-

- A smart platform that helps companies make better shipping and cargo booking decisions.
- Provides freight cost forecasts for upcoming days and weeks.
- Suggests the most suitable vessel based on cargo quantity and port conditions.
- Allows users to compare different shipping options before making a decision.
- Shows possible risks such as fuel price changes, weather conditions, and port delays



Tech Stack



Core Intelligence Layer

1. **Freight Forecasting** - Prophet + XGBoost + LSTM
2. **Vessel optimization** - Constraint filtering + ranking
3. **Voyage optimization** - Using dynamic programming for cost and idle time calculations
4. **Scenario analysis** - Fuel, weather, congestion, deadlines

Industry Validation

Maritime Dependence

95% of India's External Trade Volume Moves by Sea
[Ministry of Ports, Shipping and Waterways](#)

Cargo Scale

~855 MT of Cargo handled by major ports
[Press Release FY 2024-25](#)

Operational Efficiency

49.5 hrs of vessel turn around
[Press Release Page](#)

Optimization Opportunity

16.5% time is spent idle
[Press Release FY 2024-25](#)

FEASIBILITY

Can we build it?

- Data – Freight, fuel, commodity and port data available
- Analysis – Market trend and cost analysis
- Optimization – Vessel, port and voyage planning
- Deployment – Web-based decision support platform

VIABILITY

Why will industry adopt it?

- Lower Costs – Better vessel and booking decisions
- Faster Planning – Reduce manual analysis effort
- Reduced Risk – Evaluate congestion, weather and fuel impact
- Scalable – Applicable across ports, routes and commodities

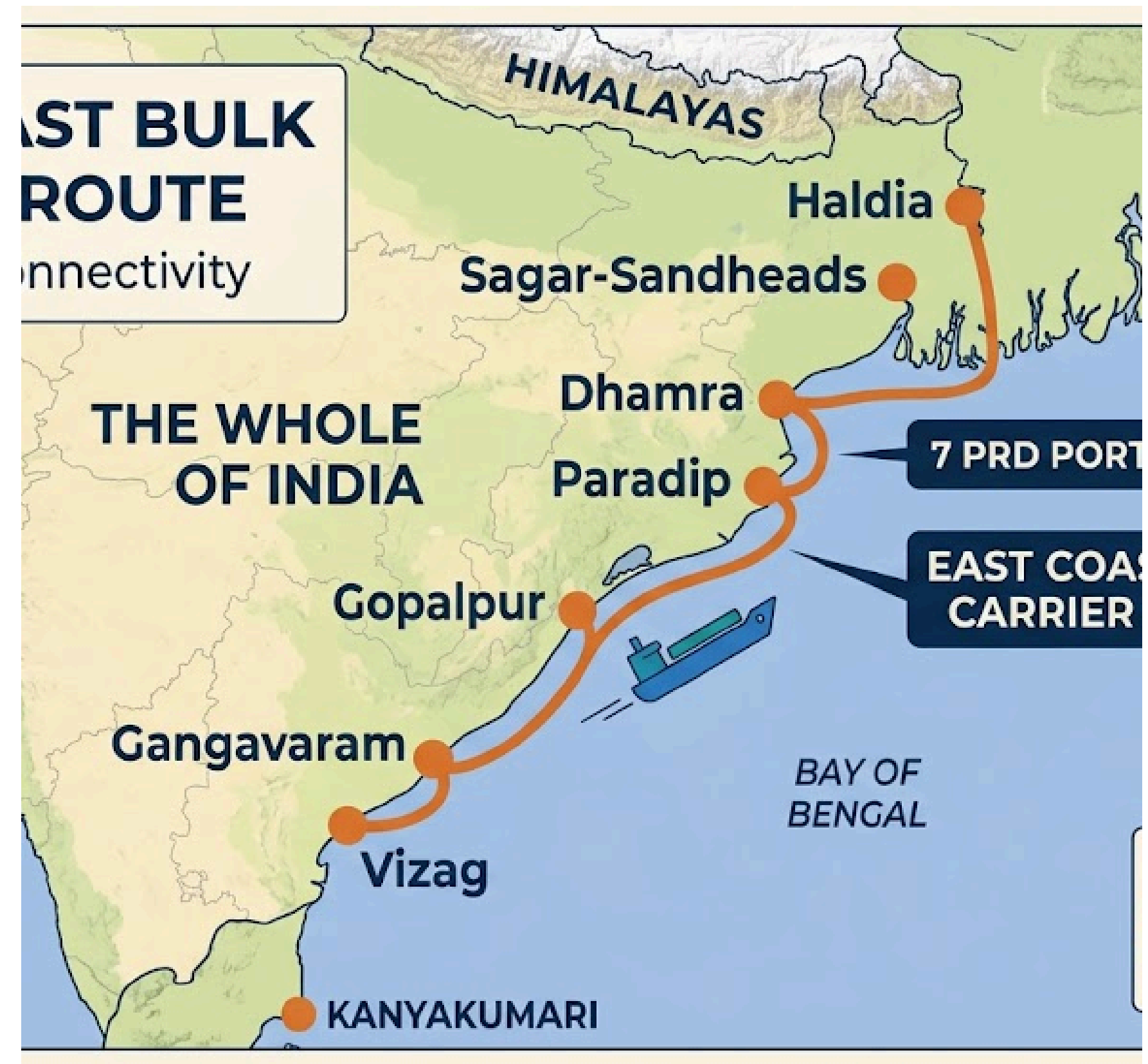
The scale of opportunity

855 MT, 95% trade volume, 1.59 BT, and the 96 to 49.5
hrs / 16.3% idle stat

"MoPSW, FY 2024-25 - decade-long national trend"

What Navora targets to deliver you?

- Lower fuel consumption and freight costs
- Reduced demurrage and vessel time
- Faster decisions
- Improved cargo flow
- Reduced operational uncertainty



Research & References

SIH 2026 Problem Statement – SIH26006

[SIH26006 Problem Statement](#)

Ministry of Ports, Shipping & Waterways (MoPSW) – Annual Reports

[MoPSW Annual Reports](#)

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Visakhapatnam Port Authority – Annual Reports

[Visakhapatnam Port Reports](#)

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Baltic Exchange – Baltic Dry Index (BDI)

[Baltic Dry Index Data](#)